

frequency analysis Equations

HWS Delayed, due after Break

Dominant pole amplifier

$$a(jf) = a_0 / (1 + jf/f_b)$$

$a_0 \rightarrow$ DC open loop gain.

$f_b \rightarrow$ open loop bandwidth

$$T(jf) = a(jf) \beta(jf)$$

$\beta(jf) \rightarrow$ feedback factor $= -\frac{V_o(jf)}{V_T}$

for resistive cases $\rightarrow \beta(jf) \rightarrow \beta$

Closed loop gain:

$$A(jf) = A_0 / (1 + jf/f_B)$$

$A_0 \rightarrow$ DC closed loop gain $= A_{ideal} / (1 + a_0 \beta)$

$f_B \rightarrow$ closed-loop bandwidth $A_0 \approx A_{ideal}$

GBWP $\rightarrow$ Gain Bandwidth Product

Open loop: $a_0 f_b = f_t$

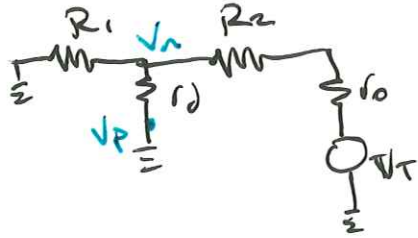
f_t occurs when $|a(jf_t)| = 1$

Closed loop: $|A_0|/f_B = f_T$

f_T occurs when $|A(jf_T)| = 1$

$$\text{where } f_B = f_b (1 + a_0 \beta) \approx f_b a_0 \beta = f_t \beta$$

Non-Inv. Amp



$$V_o = V_p - V_n = -V_n = \frac{-R_1 || r_o}{R_1 || r_o + R_2 + r_o} V_T$$

if $r_i \rightarrow \infty$, $r_o \rightarrow 0$

$$V_o = -\frac{R_1}{R_1 + R_2} V_T \rightarrow \text{~~Feedback~~ } \beta(jf) = -\frac{V_o(jf)}{V_T} = \frac{R_1}{R_1 + R_2}$$

for this case, we know

that

$$b = \frac{R_1}{R_1 + R_2} = \beta$$

$$A_{ideal} = \frac{R_1 + R_2}{R_1} = 1 + \frac{R_2}{R_1} = \frac{1}{\beta}$$

lets assume $a_o \beta$ very large
such that $A_{ideal} = A_o$

$$T(jf) = a(jf) \frac{R_1}{R_1 + R_2}$$

Aideal, not A_o

$$A(jf) = \frac{\left(1 + \frac{R_2}{R_1}\right)}{1 + \left(1 + \frac{R_2}{R_1}\right)(1 + jf/f_b)/a_o}$$

$$f_B \approx f_b a_o \beta = f_t \frac{R_1}{R_1 + R_2}$$

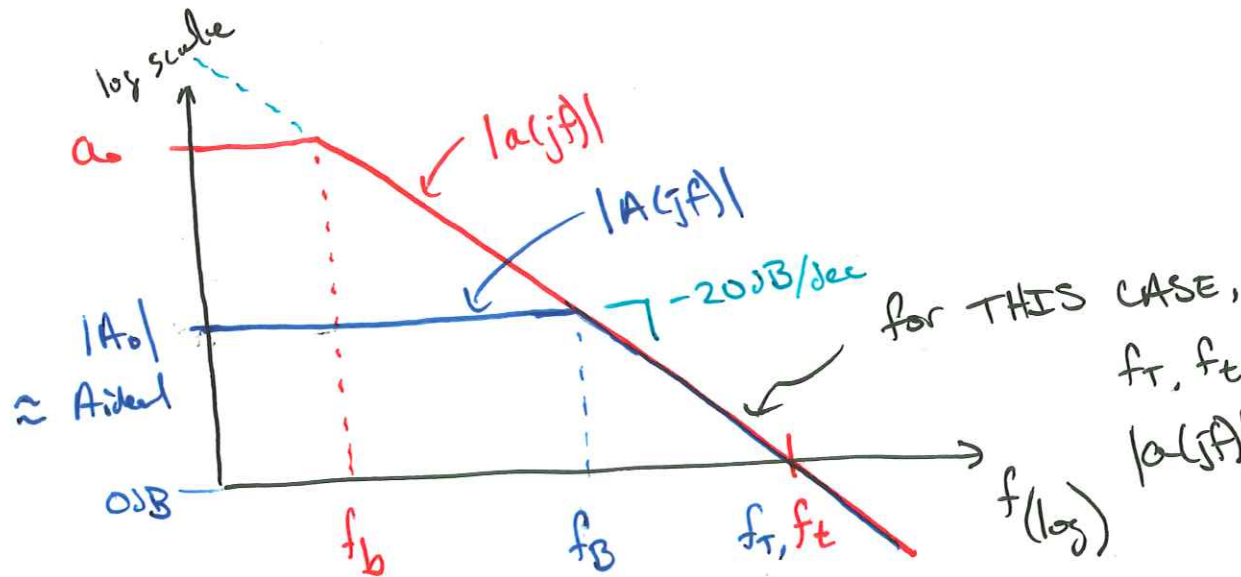
A_o, not Aideal

$$A(jf) = \frac{\left(1 + \frac{R_2}{R_1}\right)}{1 + jf/f_B}$$

$$f_T = |A_o|/f_B = \underbrace{\left(\frac{A_{ideal}}{1 + 1/a_o \beta}\right)}_{1/\beta / (1 + 1/a_o \beta) \rightarrow 0} f_b \underbrace{(1 + a_o \beta)}_{\approx a_o \beta} \approx a_o f_b = f_t$$

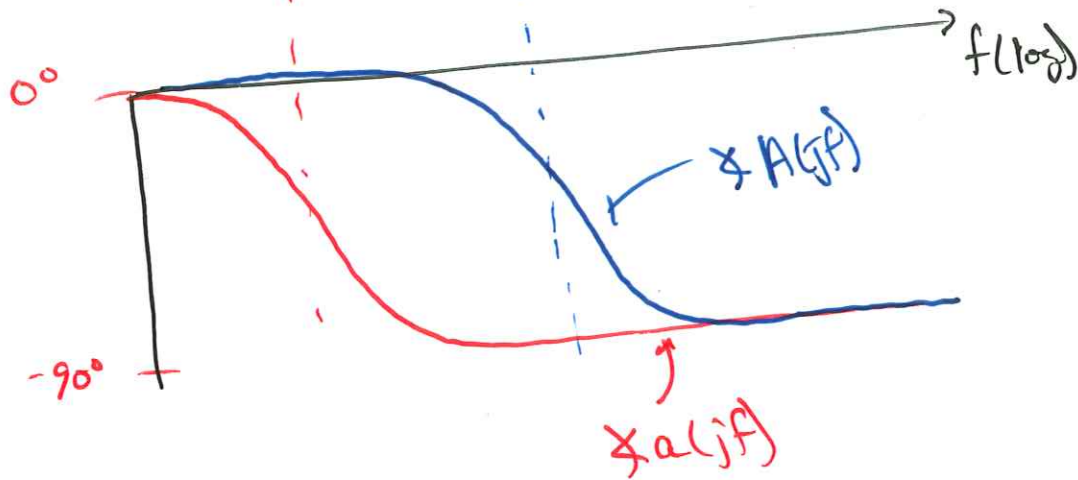
$$1/\beta / (1 + 1/a_o \beta) \rightarrow 0$$

$|a(jf)|$ & $|A(jf)|$ for non-inv. Amp.

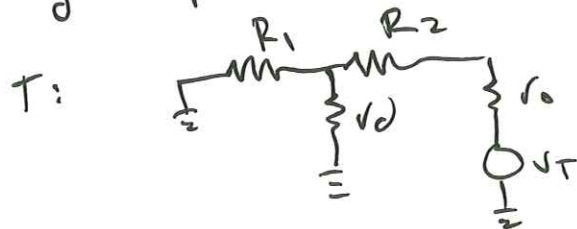


for THIS CASE,
 f_T, f_t are equal, therefore
 $|a(jf)|$ & $|A(jf)|$ decays are
 colinear.

increasing gain will result in lower bandwidth!



Inverting Amplifier



Same T as in non-inv. case!

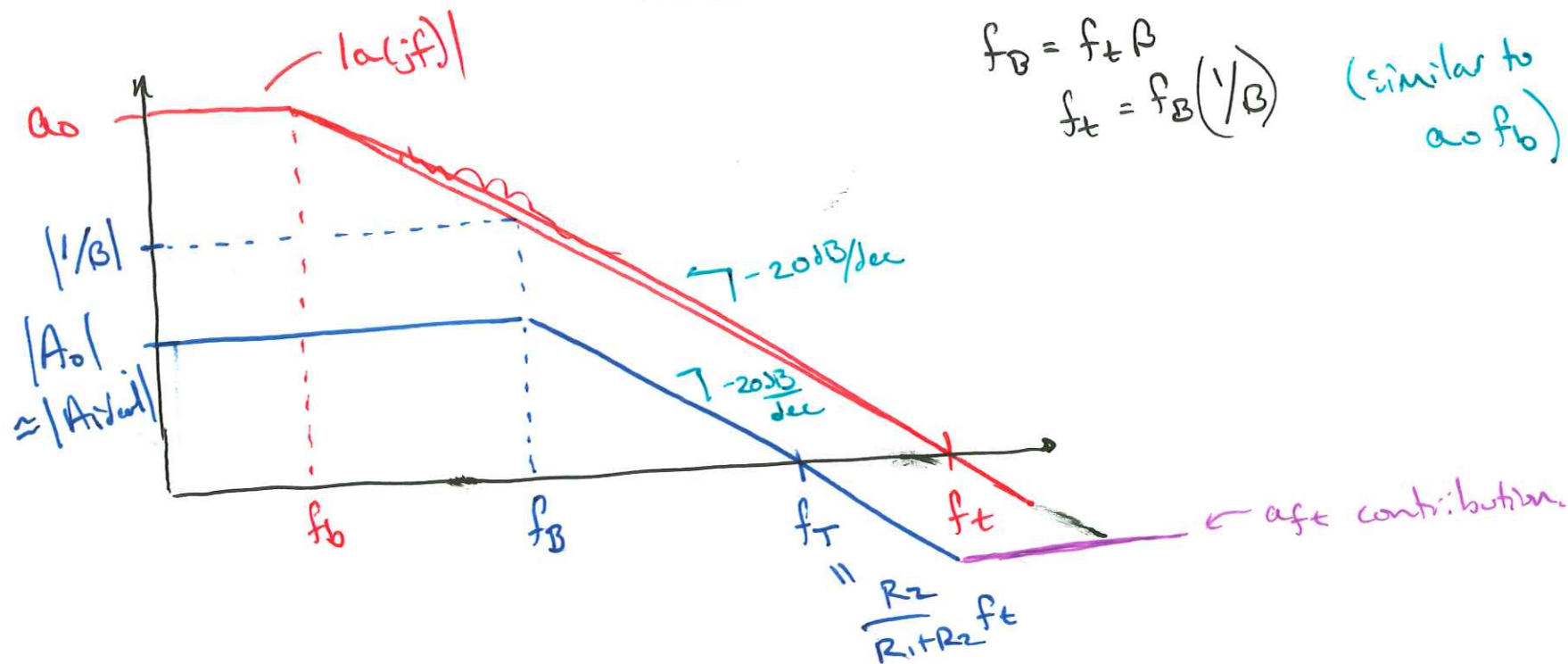
$$\therefore \beta \approx \frac{R_1}{R_1 + R_2}$$

for this case:

$$A_{ideal} = \frac{1}{b} = -\frac{R_2}{R_1} \neq \frac{1}{\beta}$$

$$f_T = |A_o| f_B = \left| \left(\frac{A_{ideal}}{1 + 1/a_o \beta} \right) \right| f_B (1 + a_o \beta) = \frac{R_2}{R_1 + R_2} a_o f_B = \frac{R_2}{R_1 + R_2} f_t$$

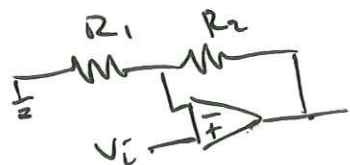
$\approx a_o \beta = a_o \frac{R_1}{R_1 + R_2}$



Impedances

$$R_i, R_o \rightarrow Z_i(jf), Z_o(jf)$$

R_i for non-inv. Amp. , $r_d \neq \infty$, $r_o \neq 0$, $r_d \gg R_1, R_2$, $r_o \ll R_1, R_2$



$$r_o \approx r_d$$

$$T_{sc}(jf) = \frac{R_1}{R_1 + R_2} a(jf)$$

$$T_{oc}(jf): V_p = V_n \Rightarrow V_d = 0 \rightarrow T_{oc}(jf) = 0$$

$$Z_i(jf) = r_o \left(\frac{1 + T_{sc}}{1 + T_{oc}} \right) \approx r_o \left(1 + \frac{R_1}{R_1 + R_2} a(jf) \right) = r_o \left(1 + \frac{a_o \beta}{1 + jf/f_b} \right)$$

$\rightarrow$ modify to be in form zeros
poles ...

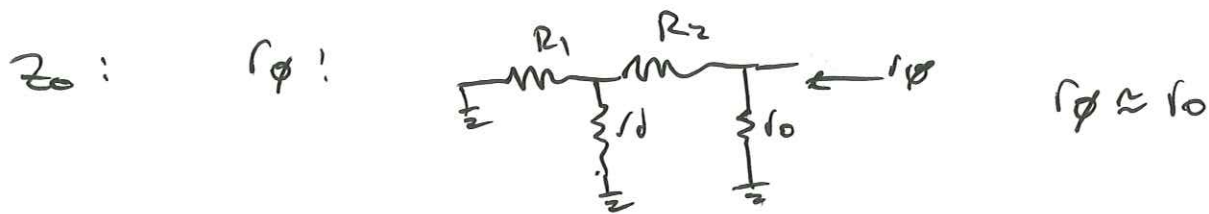
$$Z_i(jf) = r_o \left[\frac{1 + jf/f_b + a_o \beta}{1 + jf/f_b} \right] = r_o (1 + a_o \beta) \left(\frac{1 + jf/f_b (1 + a_o \beta)}{1 + jf/f_b} \right) = r_o (1 + a_o \beta) \left(\frac{1 + jf/f_B}{1 + jf/f_b} \right)$$

$$f \ll f_b \ll f_B \rightarrow Z_i(f \rightarrow 0) = r_o (1 + a_o \beta) \rightarrow R_i$$

$$f \gg f_B \gg f_b \rightarrow Z_i(f \rightarrow \infty) = r_o (1 + a_o \beta) \frac{jf/f_B}{jf/f_b} = r_o (1 + a_o \beta) \frac{f_b}{f_B}$$

$$f_B \approx a_o \beta f_b = (1 + a_o \beta) f_b$$

$$Z_i(f \rightarrow \infty) \approx r_o \left(\frac{1 + a_o \beta}{(1 + a_o \beta)} \right) \frac{f_b}{f_b} = r_o$$



~~Previous~~ T_{sc} : Output shorted to ground $\rightarrow v_n = 0, v_p = 0 \rightarrow T_{sc} = 0$

T_{oc} : Same as previous T

$$T_{oc} = \frac{R_1}{R_1 + R_2} a(jf) = \beta a(jf)$$

$$z_o \approx r_o \left(\frac{1 + 0}{1 + a\beta / (1 + jf/f_b)} \right) = r_o \left[\frac{1 + jf/f_b}{1 + jf/f_b + a\beta} \right]$$

$$z_o = \frac{r_o}{1 + a\beta} \left[\frac{1 + jf/f_b}{1 + jf/f_b (1 + a\beta)} \right] = \frac{r_o}{1 + a\beta} \left[\frac{1 + jf/f_b}{1 + jf/f_b} \right] \quad \text{--- } f_b$$

$$f \rightarrow 0 \quad z_o = r_o / (1 + a\beta)$$

$$f \rightarrow \infty: z_o = \frac{r_o}{1 + a\beta} \frac{f_b}{f_b} = r_o$$

Impedance Plots :

